

DS Unit 3 Question Bank — Trees & Binary Search Trees

Data Structures (DS) · Unit 3 · Trees & Binary Search Trees · 2025

Important and frequently asked questions, grouped by marks.

Part A — Short answer (2 marks)

1. Define Binary Tree Traversals. Derive or justify the key properties involved.
2. Write short notes on Binary Search Trees.
3. Write short notes on AVL Trees.
4. Illustrate Heaps with a worked example.
5. Explain B-Trees with a neat diagram and a suitable example.
6. Compare Threaded Binary Trees with B-Trees. When would you prefer one over the other?

Part B — Long answer (10 marks)

1. Write short notes on Binary Tree Traversals.
2. Compare Binary Search Trees with Threaded Binary Trees. When would you prefer one over the other?
3. Write short notes on AVL Trees.
4. Define Heaps. Derive or justify the key properties involved.
5. What is meant by B-Trees? Discuss its advantages and limitations.
6. Illustrate Threaded Binary Trees with a worked example.
7. Write short notes on Binary Tree Traversals.
8. Discuss the role of Binary Search Trees in Trees & Binary Search Trees.
9. Explain AVL Trees with a neat diagram and a suitable example.
10. Define Heaps. Derive or justify the key properties involved.
11. Discuss the role of B-Trees in Trees & Binary Search Trees.
12. Compare Threaded Binary Trees with B-Trees. When would you prefer one over the other?